```
c = ax[0].imshow(
    predictive_mean.detach().reshape(101, 101).transpose(-1, -2),
    origin="lower",
    extent=[0, 1, 0, 1],
) ❶

c = ax[1].imshow(
    predictive_stddev.detach().reshape(101, 101).transpose(-1, -2),
    origin="lower",
    extent=[0, 1, 0, 1],
) ❷
plt.colorbar(c, ax=ax[1])
```

❶ Heat map for the predictive mean.

❷ Heat map for the predictive standard deviation.

This will create plots similar to those in Figure A.1.

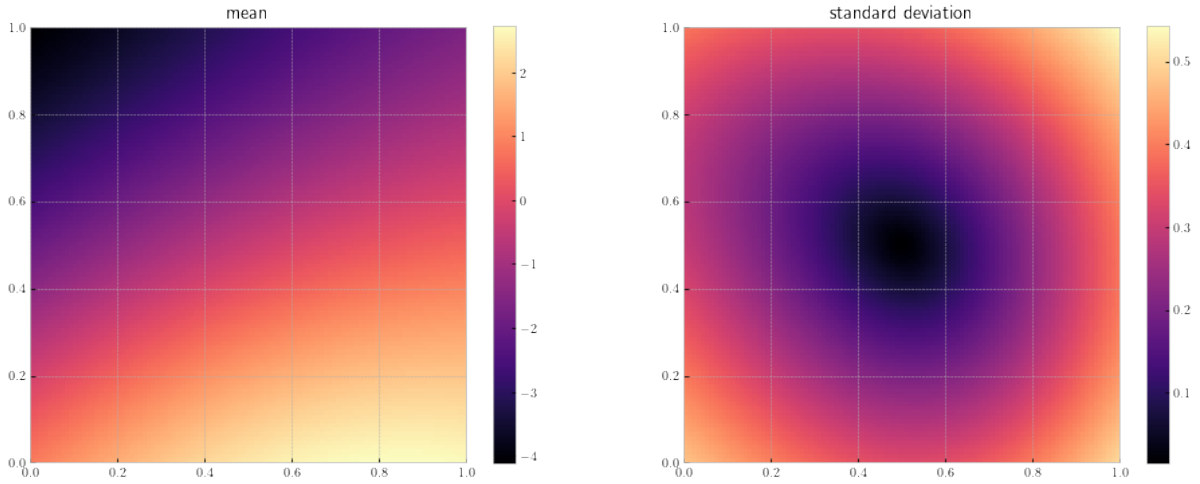


Figure A.1 Predictions made by a GP when trained on the data set in Table A.1.

NOTE **If you are using a different GP implementation, it's entirely possible to produce heat maps that are slightly different from those in Figure A.1. As long as the general trend of the heat maps is the same, your solution is correct.**

A.2 Chapter 3: Incorporating prior knowledge with the mean and covariance functions

This exercise provides a practice for implementing a GP model with ARD (automatic relevance determination). To do this, we will create an objective function that varies along an axis more than it does along another axis. We will then train a GP model, with and without ARD, on data points from this function and compare the learned length scale values.

There are multiple steps:

1. Implement the following two-dimensional function in Python using PyTorch:

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